

# Cayley-Hamilton Theorem

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# Learning Outcomes

At the end of this lecture, you should be able to:

- Understand and apply the Cayley-Hamilton Theorem.
- Use the Cayley-Hamilton Theorem to find the powers and inverse of a matrix.

# Cayley-Hamilton Theorem

## Cayley-Hamilton Theorem:

Every matrix is a zero of its characteristics polynomial. That is,  
 $A \in M_n(\mathbb{R})$ ,

$$p(\lambda) = \det(A - \lambda I_n) = 0 \Rightarrow \underline{p(A) = 0}.$$

Example:

Verify Cayley-Hamilton Theorem for  $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ .

# Application of Cayley-Hamilton Theorem

Finding the power of matrix,  $A^r$ ,  $r \geq 2$ .

Find the powers of the matrix  $A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$  (i.e. find  $A^r$ , where  $r \geq 2$ ).

$$A^3 =$$

$$A^4 =$$

In general,  $A^r = a_0 I_2 + a_1 A$ ,  $r \geq 2$ .

How to find  $a_0$  and  $a_1$ ?

# Application of Cayley-Hamilton Theorem

Finding the power of matrix,  $A^r$ ,  $r \geq 2$ .

Find the powers of the matrix  $A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$  (i.e find  $A^r$ , where  $r \geq 2$ ).

Note that each of the eigenvalues  $\lambda$  of  $A$  also satisfies the equation  $A^r = a_0 I_2 + a_1 A$ ,  $r \geq 2$ , i.e.,  $\lambda^r = a_0 + a_1 \lambda$ ,  $r \geq 2$ .

The eigenvalues of  $A$  are 2 and 5.

# Application of Cayley-Hamilton Theorem

Finding the power of matrix,  $A^r$ ,  $r \geq 2$ .

Example:

If  $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ . Determine  $A^5$  and  $A^r$ ,  $r \geq 2$ . Hence, evaluate  $A^4 - 2A^3 + A - I$ .

# Application of Cayley-Hamilton Theorem

Finding the power of matrix,  $A^r, r \geq 2$ .

What if there is eigenvalue with multiplicity more than 1?

Recall: Multiple Root Theorem:

Let  $k$  be a positive integer. Then  $a$  is a root of multiplicity  $k$  of a polynomial  $P(x)$  if and only if

$$P(a) = P'(a) = \dots = P^{(k-1)}(a) = 0 \text{ and } P^{(k)}(a) \neq 0.$$

# Application of Cayley-Hamilton Theorem

Finding the power of matrix,  $A^r$ ,  $r \geq 2$ .

Example:

If  $A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$ . Determine  $A^5$  and  $A^r$ ,  $r \geq 2$ . Hence, evaluate  $A^4 - 2A^3 + A - I$ .



# Application of Cayley-Hamilton Theorem

Finding the inverse of matrix,  $A^{-1}$ .

Example:

Let  $A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ . Use the Cayley-Hamilton Theorem to find the inverse of  $A$ .

# Application of Cayley-Hamilton Theorem

Example:

Let  $A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$ .

- (a) Find the characteristic polynomial of  $A$ .
- (b) Show that  $A$  is a root of its characteristic polynomial.
- (c) If  $A$  is invertible, find  $A^{-1}$  by using the Cayley-Hamilton theorem.

# Application of Cayley-Hamilton Theorem

Example:

Let  $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ . Determine  $A^5$ ,  $A^r$ ,  $r \geq 2$  and if possible,  $A^{-1}$ .

# Summary

Let  $A \in M_n(\mathbb{R})$ . If  $A$  is invertible, then

- $A$  is nonsingular.
- The homogeneous linear system  $AX = 0$  has only the trivial solution.
- The linear system  $AX = B$  has a unique solution for each  $B \in M_{n,1}(\mathbb{R})$ .
- The reduced row echelon form of  $A$  is  $I_n$ .
- $A \stackrel{r}{\sim} I_n$ .
- $A$  can be expressed as a product of elementary matrices,  $E_1 E_2 \dots E_k$ .
- $\det A \neq 0$ .
- $\text{adj} A$  is invertible.
- The row vectors of  $A$  are linearly independent in  $\mathbb{R}^n$ .
- The column vectors of  $A$  are linearly independent in  $M_{n,1}(\mathbb{R})$ .
- $\rho(A) = n$ .
- 0 is not an eigenvalue of  $A$ .
- The constant term of the characteristic polynomial of  $A$  is nonzero.

# Summary

Ways to find the inverse of an invertible matrix  $A \in M_n(\mathbb{R})$ .

- The definition  $AA^{-1} = A^{-1}A = I_n$ .
- Matrix inversion algorithm  $[A|I_n] \longrightarrow [I_n|A^{-1}]$ .
- With the adjoint matrix of  $A$ ,  $A^{-1} = \frac{\text{adj}A}{|A|}$ .
- Cayley-Hamilton Theorem.